

# MEDS-S1 Project Catalogue

31 projects · state your preferences

UET Lahore · Department of Electrical Engineering · August 2026

Every project on the following pages is a real component of MEDS-S1, with a specification, a testbench and a merge gate. There is no practice work in this catalogue. Read the pages that interest you, talk to your mentor about what a project actually involves, and send **three ranked preferences with one line of reasoning each**.

16

Mentee projects  
1–1.5 months

8

Mentor projects  
2–3 months

7

Graduate RA projects  
~3 months

MEDS-S1 is an open RISC-V SoC platform whose purpose is to be attached to. Apache-2.0.

Companion documents: specs/MEDS-S1-SPECIFICATION.md · specs/INTERFACES.md · specs/SCOPE\_CONTRACT.md

# How to use this catalogue

Read this page before you read the projects

**This is a menu, not an assignment.** There are more projects here than we can staff this cycle. Priority markings say which ones start now: **P0** begins immediately and is on the critical path, **P1** starts this cycle, **P2** is queued for the next one. A project you want that is marked P2 is still worth naming — it tells us where to grow.

## WHAT EVERY PAGE TELLS YOU

- **Tier** — who the project is sized for: mentee, mentor or graduate RA. It is a size, not a ranking. The hardest verification project in this catalogue is a mentee project.
- **Team** and **duration** — how many people, and how long. Durations assume part-time work alongside coursework.
- **Literature review** — every project starts with reading and produces a written memo or note before implementation begins. This is not a warm-up exercise; the note is a deliverable and it is reviewed. Design review before RTL, always.
- **Definition of done** — binary criteria. A project is not finished because the code exists.

## HOW TO STATE A PREFERENCE

- Send **three**, ranked, to your mentor. Mentors consolidate and pass to the graduate RAs.
- Give **one line of reasoning** per choice. Where two people want the same project, the reasoning decides it — so write it properly.
- A useful preference: *"M-05, the PMA testbench. I want to learn how the memory map is enforced and I wrote SystemVerilog testbenches during training."*
- A useless preference: *"anything", "whatever is easiest", "the most impressive one".*

## THE CONTRIBUTION LADDER

- **T0 — Contributor.** Add a peripheral via soc.yaml, write a driver, write a directed testbench, add a benchmark. Entry: rv-workshop completed.
- **T1 — Implementer.** Implement a module against a frozen spec. Entry: two merged T0 PRs.
- **T2 — Owner.** Own a module and its spec; review T1 work. Entry: one T1 module delivered *and verified*.
- **T3 — Architect.** Own an interface. Only these people may change INTERFACES.md.

Everyone starts at T0, including people who think they should not. The tiers are a ladder, not a label — a mentee who delivers M-05 well is a stronger candidate for a T-tier project next cycle than someone who was handed one this time.

## HOW 16 + 8 + 7 PROJECTS MAP ONTO THE PEOPLE WE HAVE

- **22 mentees, 16 projects, teams of 1–2.** That absorbs everyone with room to spare, which is the point — preferences only mean something when there is genuine choice.
- **8 mentors, 8 projects.** Four mentors are committed to MEDS-V as their final year project, so expect three or four of these to be staffed this cycle and the rest to be queued or co-owned. Name the one you want anyway.
- **3 graduate RAs, 7 projects.** Clearly a menu rather than a roster. The P0 entries — R-01, R-04, R-05 and R-06 — start first, because Phase 1 does not begin without them.

**Two rules that are not negotiable.** Design review before RTL — a module owner presents the spec and the testbench plan, and the review approves the *spec*, not the code. And blockers are raised the same day: a mentee blocked for a week has lost a sixth of a semester.

# Index — Mentee projects

16 projects · Training completion projects — small, bounded, 1 to 1.5 months

| ID   | PROJECT                                                               | TRACK           | TEAM | DURATION | PRIORITY | WP         |
|------|-----------------------------------------------------------------------|-----------------|------|----------|----------|------------|
| M-01 | UART Peripheral Integration and HAL Driver                            | Fabric & Periph | 2    | 6 weeks  | P0       | WP9, WP14  |
| M-02 | SPI Master Peripheral, Driver and Flash Boot Path                     | Fabric & Periph | 2    | 6 weeks  | P1       | WP9, WP14  |
| M-03 | GPIO and Timer Peripherals with Drivers                               | Fabric & Periph | 2    | 5 weeks  | P1       | WP9, WP14  |
| M-04 | Boot ROM and Second-Stage Loader                                      | Software & BSP  | 2    | 6 weeks  | P1       | WP9, WP14  |
| M-05 | PMA Check Unit — Directed Testbench                                   | Verification    | 1–2  | 5 weeks  | P1       | WP13       |
| M-06 | ALU and Forwarding Network — Unit Testbench and Coverage              | Verification    | 1–2  | 5 weeks  | P1       | WP13       |
| M-07 | CoreMark and Dhrystone Benchmark Harness                              | Software & BSP  | 2    | 5 weeks  | P1       | WP19       |
| M-08 | Embench-IoT Port and Automated Regression                             | Software & BSP  | 2    | 6 weeks  | P2       | WP19       |
| M-09 | libs1_perf — the Performance Measurement Library                      | Software & BSP  | 2    | 6 weeks  | P0       | WP14, WP19 |
| M-10 | Lint, Coding Standard and CI Documentation Gates                      | Docs & Research | 1–2  | 4 weeks  | P0       | WP1, WP22  |
| M-11 | RISC-V Architectural Compatibility Tests (ACT) — Triage and Reporting | Verification    | 2    | 6 weeks  | P0       | WP12       |
| M-12 | Xilinx IP Survey and Evaluation for KC705 Synthesis                   | FPGA & Tools    | 2    | 6 weeks  | P0       | WP16       |
| M-13 | Module Documentation and Interface-Contract Sweep                     | Docs & Research | 1–2  | 4 weeks  | P1       | WP22       |
| M-14 | Literature Review — Open-Source RISC-V Core Microarchitectures        | Docs & Research | 2–3  | 4 weeks  | P0       | WP0        |
| M-15 | Literature Review — Accelerator Coupling and Measurement Methodology  | Docs & Research | 2    | 5 weeks  | P1       | WP0, WP17  |
| M-16 | The Bridge Exercise — Port a Workshop Core into the MEDS-S1 Harness   | Verification    | 2    | 6 weeks  | P1       | WP13       |

# Index — Mentor projects

8 projects · Larger components — one owner, one spec, one testbench

| ID   | PROJECT                                                      | TRACK           | TEAM | DURATION | PRIORITY | WP   |
|------|--------------------------------------------------------------|-----------------|------|----------|----------|------|
| T-01 | Core Frontend — PC, Branch Prediction and Fetch Interface    | Core            | 2    | 10 weeks | P0       | WP2  |
| T-02 | Core Backend — Decode, Register File, ALU and Hazard Control | Core            | 2–3  | 12 weeks | P0       | WP3  |
| T-03 | CSR File, Trap Architecture and Performance Counters         | Core            | 2    | 12 weeks | P0       | WP4  |
| T-04 | AXI4 Fabric — Crossbar, Width Adaptation and Address Decode  | Fabric & Periph | 2    | 10 weeks | P0       | WP8  |
| T-05 | CLINT and PLIC — the Interrupt Subsystem                     | Fabric & Periph | 2    | 8 weeks  | P1       | WP9  |
| T-06 | Board Support Package — crt0, newlib, HAL and make run       | Software & BSP  | 2    | 10 weeks | P0       | WP14 |
| T-07 | RISCOF, Architectural Tests and Co-simulation in CI          | Verification    | 2    | 8 weeks  | P0       | WP12 |
| T-08 | KC705 Board Port — DDR3 Bring-Up and Timing Closure          | FPGA & Tools    | 2–3  | 12 weeks | P1       | WP16 |

# Index — Graduate RA projects

7 projects · Extra-large components — on the critical path, or an interface

| ID   | PROJECT                                                      | TRACK        | TEAM | DURATION | PRIORITY           | WP   |
|------|--------------------------------------------------------------|--------------|------|----------|--------------------|------|
| R-01 | Completion Buffer, Retire Logic and the MXIF Port            | Core         | 1–2  | 12 weeks | P0 — critical path | WP5  |
| R-02 | Load/Store Unit, Store Buffer, PMP and Atomics               | Memory       | 1–2  | 12 weeks | P0                 | WP6  |
| R-03 | Instruction and Data Caches, Zicbom and the SRAM Wrapper     | Memory       | 1–2  | 12 weeks | P1                 | WP7  |
| R-04 | The SoC Generator — soc.yaml to RTL, Linker, Headers and DTS | FPGA & Tools | 1–2  | 12 weeks | P0                 | WP10 |
| R-05 | RVFI Trace Port and the Spike Co-simulation Harness          | Verification | 1–2  | 10 weeks | P0                 | WP11 |
| R-06 | Debug Module, JTAG DTM, OpenOCD and Semihosting              | FPGA & Tools | 1–2  | 10 weeks | P0                 | WP15 |
| R-07 | Accelerator Socket and the Conformance Testbenches           | Memory       | 1–2  | 10 weeks | P1                 | WP17 |

## UART Peripheral Integration and HAL Driver

Mentee Fabric & Periph

|           |          |          |                                     |
|-----------|----------|----------|-------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                       |
| 2         | 6 weeks  | P0       | rv-workshop; basic SystemVerilog; C |

Bring a reused UART onto the AXI4-Lite peripheral subtree, give it an entry in soc.yaml, and write the HAL driver and printf backend the whole lab will use for the next four years. This is the smallest complete slice of the platform: an IP, a bus wrapper, a testbench, a driver and a C program that prints.

### WHAT YOU WILL BUILD

- Evaluate the OpenTitan and PULP UART IP; choose one and record the licence
- AXI4-Lite wrapper plus a soc.yaml entry — base address, IRQ line, PMA attributes
- Directed testbench: TX, RX, framing error, FIFO full and empty, IRQ assert and clear
- HAL driver: uart\_init, uart\_putc, uart\_getc, uart\_write
- newlib \_write hook so printf() works over UART

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- OpenTitan UART and PULP APB UART specifications
- INTERFACES.md P1–P5 — PMA rules for device regions
- Deliverable: a 2-page memo comparing the two IPs on features, FIFO depth, IRQ model, licence and RTL size, ending in a recommendation

### DELIVERABLES

- RTL wrapper in rtl/peripherals/ and the soc.yaml entry
- Unit testbench running in CI
- HAL driver and header in sw/bsp/
- IP comparison memo, and a module README stating the interface contract

### DEFINITION OF DONE

- printf("Hello World\n") from C runs on the Verilator board
- Testbench green in CI; Verible lint clean with no waivers
- README.md present in the module directory (NFR-7)

### MILESTONES

- W1** Literature review and IP selection memo
- W2–3** AXI4-Lite wrapper and soc.yaml entry
- W4** Directed testbench
- W5** HAL driver and printf backend
- W6** Integration, documentation, PR review

**Start here:** Spec §24 (peripherals), SCOPE\_CONTRACT §5 (reuse policy), PROJECTS.md WP9/WP14

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Fabric & Periph

|           |          |          |                                        |
|-----------|----------|----------|----------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                          |
| 2         | 6 weeks  | P1       | rv-workshop; SystemVerilog; a little C |

## WHAT YOU WILL BUILD

- ## LITERATURE REVIEW & THE NOTE IT PRODUCES

- ## DELIVERABLES

- SPI RTL wrapper, soc.yaml entry, unit testbench in CI
- HAL driver plus the flash-read example application
- Module README with the register map and the interface contract

## DEFINITION OF DONE

- Testbench reads a known pattern from the behavioural flash model, in CI
- Driver API agreed and used by the M-04 boot-loader project
- Lint clean; README present

## MILESTONES

- |             |                                                        |
|-------------|--------------------------------------------------------|
| <b>W1</b>   | Reading, IP selection, interface note agreed with M-04 |
| <b>W2–3</b> | RTL wrapper and register window                        |
| <b>W4</b>   | Testbench and flash model                              |
| <b>W5</b>   | HAL driver                                             |
| <b>W6</b>   | Example application and documentation                  |

NOTES · QUESTIONS FOR YOUR MENTOR

## GPIO and Timer Peripherals with Drivers

Mentee Fabric & Periph

|           |          |          |                                   |
|-----------|----------|----------|-----------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                     |
| 2         | 5 weeks  | P1       | rv-workshop; SystemVerilog basics |

Deliver the two peripherals every board bring-up depends on. GPIO is how the first LED blinks and how bring-up step 1 is passed; the timer is how software measures anything before the performance counters are trusted. Small, well-defined, and on the critical path of the first FPGA milestone.

### WHAT YOU WILL BUILD

- GPIO: N-bit configurable direction, output, input synchroniser, per-pin interrupt
- Timer: free-running counter, compare registers, periodic and one-shot interrupt
- Both with AXI4-Lite register windows and soc.yaml entries
- Directed testbenches for both, including the interrupt paths
- HAL drivers and a blink plus periodic-interrupt example

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The CLINT mtime/mtimecmp model in the RISC-V privileged specification
- Read spec §24 and note how a peripheral IRQ reaches the core through the PLIC
- Deliverable: a half-page note on why the timer here is not the same thing as CLINT mtime, and when software should use each

### DELIVERABLES

- Two RTL peripherals, two soc.yaml entries, two unit testbenches in CI
- Two HAL drivers plus the example application
- Module READMEs for both

### DEFINITION OF DONE

- LED blinks on the Verilator board from C, driven by the timer interrupt
- Both testbenches green in CI; lint clean
- Interrupt paths exercised in simulation, not just the register reads

### MILESTONES

- W1** Reading and register-map design, reviewed before RTL
- W2** GPIO RTL and testbench
- W3** Timer RTL and testbench
- W4** HAL drivers
- W5** Example application and documentation

**Start here:** Spec §24, RISC-V Privileged Spec (CLINT), PROJECTS.md WP9

NOTES · QUESTIONS FOR YOUR MENTOR

## Boot ROM and Second-Stage Loader

Mentee

Software &amp; BSP

|           |          |          |                                          |
|-----------|----------|----------|------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                            |
| 2         | 6 weeks  | P1       | C, assembly, linker scripts; rv-workshop |

Write the first code that runs on MEDS-S1. The boot ROM sets up the machine from reset and hands over to a second-stage loader that can pull an image in over UART or QSPI. Everything anybody ever runs on this platform passes through the few hundred instructions you write here.

## WHAT YOU WILL BUILD

- Reset vector, stack setup, BSS clear, trap-vector install, jump to second stage
- A 32 KB ROM image generated into the RTL from the build, never hand-edited
- Second-stage loader: XMODEM-style receive over UART, plus a QSPI path using M-02's driver
- Platform version reporting — read `MEDS_S1_PLATFORM_VERSION` and print an identifying banner
- A boot-time self-test that checks SRAM is readable and writable before handing over

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- Read two existing boot ROMs — the CVA6 and the Ibex/OpenTitan ones — and note what each does before jumping to user code
- The generated linker script and memory map in soc.yaml
- Deliverable: a 1-page boot-flow diagram, from reset vector to main(), reviewed before any code is written

## DELIVERABLES

- crt0-adjacent boot ROM source in `sw/boot/`, plus the ROM image generation step
- Second-stage loader with UART and QSPI paths
- Boot-flow diagram and a README that documents every ROM entry point

## DEFINITION OF DONE

- A cold reset in Verilator reaches `main()` in a C program and prints the version banner
- The loader receives a small ELF over the UART model and runs it
- ROM image is build-generated; no checked-in binary blob

## MILESTONES

|             |                                   |
|-------------|-----------------------------------|
| <b>W1</b>   | Reading and the boot-flow diagram |
| <b>W2–3</b> | Boot ROM and ROM generation       |
| <b>W4</b>   | UART second-stage path            |
| <b>W5</b>   | QSPI path with M-02               |
| <b>W6</b>   | Self-test, banner, documentation  |

**Start here:** Spec §24, FR-15, SCOPE CONTRACT §2.2, PROJECTS.md WP9

NOTES · QUESTIONS FOR YOUR MENTOR

**Mentee** **Verification**

Write the testbench that proves the PMA check unit enforces every region attribute correctly. This is the module that decides whether a load is cacheable, whether a device access may be reordered, and whether an access to a hole traps — and the failure mode when it is wrong is a bug that looks like a cache bug for three weeks.

- A testbench covering all six attributes across every region in the default memory map
- Cross coverage: region  $\times$  access type  $\times$  size  $\times$  alignment
- Negative tests: access to an unmapped hole, misaligned atomic, write to a read-only region
- Ordering checks for order:strong device regions (INTERFACES.md P2)
- A self-checking scoreboard, not eyeballed waveforms

- Spec §11 (PMP and PMA checking) and the PMA table, in full
- RISC-V Privileged Spec, the physical memory attributes chapter
- Deliverable: a 1-page table of every region and the attribute set you will test, signed off by the WP6 owner before you write the testbench

- `verif/unit/tb_pma_check.sv` with a self-checking scoreboard
- A functional coverage group for the PMA cross
- A short report of the coverage achieved and any holes you could not close, with reasons

- Testbench runs in CI on every PR and takes under 60 seconds
- Every region in the memory map is exercised for every attribute
- Coverage report published; any uncovered bin has a written reason

|           |                                        |
|-----------|----------------------------------------|
| <b>W1</b> | Reading and the region/attribute table |
| <b>W2</b> | Testbench skeleton and stimulus        |
| <b>W3</b> | Scoreboard and negative tests          |
| <b>W4</b> | Coverage model                         |
| <b>W5</b> | CI integration and the coverage report |

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Verification

Prove the arithmetic and the forwarding paths are right, exhaustively where you can and by constrained-random where you cannot. Forwarding bugs are the classic pipeline defect: they pass every simple test and fail on a specific back-to-back sequence. A good testbench here saves the WP3 owner weeks.

- Exhaustive testing of the ALU operations that admit it, and directed corner cases for the rest
- A constrained-random layer generating back-to-back dependent instruction pairs and triples
- Reference model in SystemVerilog or C for result checking
- Coverage of the full EX->EX and MEM->EX forwarding matrix from spec §8.1
- The load-use stall case explicitly, since it is the one hazard that is not forwarded

- Spec §8 (hazards, forwarding and stalls) including the hazard table, in full
- One chapter of a standard architecture text on data hazards, for the vocabulary
- Deliverable: a filled-in copy of the §8.2 hazard table marking which row each of your tests covers

- `verif/unit/tb_alu.sv` and `verif/unit/tb_forwarding.sv`
- Reference model and the coverage groups
- The annotated hazard table showing test-to-hazard traceability

- Every row of the §8.2 hazard table has at least one directed test
- Forwarding matrix coverage at 100%, report published
- Both testbenches in CI and green

|           |                                               |
|-----------|-----------------------------------------------|
| <b>W1</b> | Reading and the hazard-table traceability map |
| <b>W2</b> | ALU testbench and reference model             |
| <b>W3</b> | Forwarding testbench                          |
| <b>W4</b> | Constrained-random layer                      |
| <b>W5</b> | Coverage closure and CI integration           |

NOTES · QUESTIONS FOR YOUR MENTOR

## CoreMark and Dhrystone Benchmark Harness

Mentee

Software &amp; BSP

|           |          |          |                          |
|-----------|----------|----------|--------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES            |
| 2         | 5 weeks  | P1       | C, make, shell scripting |

Stand up the two benchmarks the outside world compares cores on, and automate them so a number appears on every pull request without anybody asking. NFR-2 sets a target of 1.0 CoreMark/MHz; you build the thing that says whether we met it, and that notices the day we stop meeting it.

### WHAT YOU WILL BUILD

- Port CoreMark and Dhrystone to the BSP, with correct timing via mcycle
- A results parser producing machine-readable output, not just console text
- CI integration reporting CoreMark/MHz and DMIPS/MHz on each PR
- A regression gate: a drop of more than 3% blocks the merge (NFR-10)
- A results-history file so the numbers can be plotted over the life of the project

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The CoreMark run rules — particularly what may and may not be changed, and which compiler flags are reportable
- Why Dhrystone is criticised, and what it is still useful for
- Deliverable: a 1-page note stating the exact flags, library and reporting form we will use, so our numbers stay comparable to published ones

### DELIVERABLES

- sw/benchmarks/ with both benchmarks and the run scripts
- CI job publishing the results table
- The run-rules note and a README on reproducing a number by hand

### DEFINITION OF DONE

- make bench BOARD=verilator prints a CoreMark/MHz figure
- CI publishes the number on every PR and blocks on a >3% regression
- The reported configuration and flags are documented alongside every number

### MILESTONES

- |             |                                      |
|-------------|--------------------------------------|
| <b>W1</b>   | Run rules note and toolchain flags   |
| <b>W2–3</b> | Porting both benchmarks onto the BSP |
| <b>W4</b>   | Parser and CI job                    |
| <b>W5</b>   | Regression gate and history file     |

**Start here:** Spec §32.1, NFR-2, NFR-10, PROJECTS.md WP19

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee      Software & BSP

Embench-IoT is the modern, harder-to-game embedded benchmark suite, and it is the one a reviewer will ask about. Porting all of it gives per-benchmark cycle counts that show where the core is weak — which is exactly the information a v2 microarchitecture project needs in order to be worth doing.

- Port the Embench-IoT suite to the BSP and the linker script
- Per-benchmark cycle and instruction counts collected through `libs1_perf`
- A comparison table against the published reference scores
- Nightly CI job, since the full suite is too slow for the per-PR budget
- A short analysis of the three slowest benchmarks and why they are slow

- The Embench-IoT papers and the project's stated design goals
- Spec §32.1 and §34 (comparison methodology and the required reporting table)
- Deliverable: a 2-page memo on what Embench measures that CoreMark does not, with the specific benchmarks that expose each difference

- `sw/benchmarks/embench/` with the full ported suite
- Nightly CI job and the results table in the evidence bundle format
- The analysis memo on the three slowest benchmarks

- Every benchmark in the suite builds and runs to a correct result
- Nightly job publishes a per-benchmark table
- The analysis identifies at least one concrete, measurable v2 improvement

|             |                                 |
|-------------|---------------------------------|
| <b>W1</b>   | Reading and the memo            |
| <b>W2–4</b> | Porting, benchmark by benchmark |
| <b>W5</b>   | CI job and results table        |
| <b>W6</b>   | Analysis of the slow cases      |

**Start here:** Spec §32.1, §34, PROJECTS.md WP19

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Software & BSP

Build the library every result from this platform will be measured with. If `libs1_perf` is good, a 2031 thesis is comparable to a 2027 one; if it is sloppy, none of the numbers this lab publishes can be trusted. Small in code, large in consequence.

- PERF\_BEGIN / PERF\_END region markers with negligible overhead
- A programmable event-selector API over mhpmmcounter3–15 (spec §12)
- Counter save and restore so nested regions work correctly
- A report formatter that emits the §34.1 required reporting table directly
- An overhead self-measurement, so users know the cost of measuring

- Spec §12 (performance counters) and §31 (measurement infrastructure)
- Spec §34 — the required reporting table and the rules around it
- Deliverable: a 1-page note on why kernel-only speedup is misleading and what end-to-end measurement has to include; this becomes the library's doc header

- `sw/bsp/libs1_perf/` with the API, headers and the report formatter
- A worked example measuring a small workload end to end
- The methodology note, and a README with the counter event list

- The library emits the §34.1 table for an example workload with no hand editing
- Measurement overhead is itself measured and documented
- Used by at least one benchmark project (M-07 or M-08) before sign-off

|             |                                                |
|-------------|------------------------------------------------|
| <b>W1</b>   | Reading and the methodology note               |
| <b>W2–3</b> | Counter API and region markers                 |
| <b>W4</b>   | Report formatter for the §34 table             |
| <b>W5</b>   | Overhead measurement                           |
| <b>W6</b>   | Integration with a benchmark and documentation |

NOTES · QUESTIONS FOR YOUR MENTOR

## Lint, Coding Standard and CI Documentation Gates

Mentee

Docs &amp; Research

| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                              |
|-----------|----------|----------|--------------------------------------------|
| 1-2       | 4 weeks  | P0       | Some SystemVerilog; care about consistency |

Write the rules everyone's RTL is held to, and the CI jobs that enforce them without a human having to nag. This is a small project with unusual leverage: it runs on every commit anybody makes for the next four years, and it is one of the few Phase-0 items that must land before RTL starts.

### WHAT YOU WILL BUILD

- CODING\_STANDARD.md — naming, file layout, reset style, parameter conventions
- A Verible lint configuration matching it, with a documented waiver process
- A CI job checking every module directory has a README stating its interface contract (NFR-7)
- A CI job checking SPDX headers and licence consistency on every file
- A pre-commit hook so contributors see failures before CI does

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The lowRISC and PULP SystemVerilog style guides
- Appendix D of the specification — RTL naming conventions
- Deliverable: a 1-page summary of where the two style guides disagree and which one MEDS-S1 follows in each case, with the reasoning

### DELIVERABLES

- CODING\_STANDARD.md, the Verible config, and the waiver process documented
- Two CI jobs (README check, SPDX check) and the pre-commit hook
- The style-guide comparison note

### DEFINITION OF DONE

- make ci runs lint and both doc gates and is green on the current tree
- A deliberately malformed test branch is correctly rejected by each gate
- Zero waivers in the tree without a written justification

### MILESTONES

- |           |                                                  |
|-----------|--------------------------------------------------|
| <b>W1</b> | Reading and the style comparison note            |
| <b>W2</b> | CODING_STANDARD.md and the Verible config        |
| <b>W3</b> | CI gates                                         |
| <b>W4</b> | Pre-commit hook, negative testing, documentation |

**Start here:** NFR-4, NFR-7, Appendix D, PROJECTS.md WP1/WP22

NOTES · QUESTIONS FOR YOUR MENTOR

## RISC-V Architectural Compatibility Tests (ACT) — Triage and Reporting

Mentee

Verification

|           |          |          |                                                                |
|-----------|----------|----------|----------------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                                  |
| 2         | 6 weeks  | P0       | Linux command line, Python, patience; no RTL experience needed |

Run the RISC-V International ACT suite — the architectural compatibility tests, formerly riscv-arch-test — against the reference model and, as it appears, against S1-Core. Triage every failure into a real bug, a test-harness problem or an unimplemented feature. This is mostly process and reporting work rather than design work, and it is the evidence that lets MEDS-S1 claim compatibility at all.

### WHAT YOU WILL BUILD

- A working RISCOF setup with Sail as the golden reference model
- The full ACT suite running for RV64IMAC\_Zicsr\_Zifencei\_Zicbom, config by config
- A triage log classifying every failure: core bug, harness problem, or out-of-scope feature
- A compatibility report in the form that goes into the release evidence bundle
- An issue raised, with a minimal reproducer, for every failure classified as a core bug

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The RISCOF documentation and the ACT repository structure and naming rules
- How a test signature is produced and compared, end to end
- Deliverable: a 2-page explainer on what architectural compatibility does and does not prove — this goes to the whole lab, because it is widely misunderstood

### DELIVERABLES

- The RISCOF configuration and plugins committed to the repo
- The triage log and the compatibility report
- Issues filed with reproducers; the explainer memo

### DEFINITION OF DONE

- The suite runs end to end from a single make target
- Every failure is classified, with a reason, and none are left unexplained
- The report is in evidence-bundle form (spec §28.6)

### MILESTONES

- |             |                                          |
|-------------|------------------------------------------|
| <b>W1</b>   | Reading and the explainer memo           |
| <b>W2</b>   | RISCOF and Sail installed and running    |
| <b>W3–4</b> | Full suite execution and triage          |
| <b>W5</b>   | Report and issue filing                  |
| <b>W6</b>   | Automation so it reruns from one command |

**Start here:** Spec §28.3, §28.6, FR-1, PROJECTS.md WP12; RISCOF and riscv-arch-test upstream

NOTES · QUESTIONS FOR YOUR MENTOR

## Xilinx IP Survey and Evaluation for KC705 Synthesis

Mentee

FPGA &amp; Tools

| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                     |
|-----------|----------|----------|---------------------------------------------------|
| 2         | 6 weeks  | P0       | Vivado installed; no RTL design experience needed |

Survey the Xilinx IP we will lean on when MEDS-S1 meets real silicon, and produce configuration recipes the FPGA team can use rather than rediscover. Getting MIG and the clocking right is most of what makes an FPGA bring-up take two weeks instead of two months, and none of it requires writing core RTL.

### WHAT YOU WILL BUILD

- An evaluation of MIG for DDR3 on KC705: interface width, clocking, calibration, AXI options
- Clocking Wizard, Block Memory Generator and AXI SmartConnect configurations for our topology
- ILA, VIO and JTAG-to-AXI debug cores — what each costs in area and what each is worth
- A small standalone Vivado project per IP that builds and proves the configuration works
- Resource and timing figures for each IP at our target clock, measured, not estimated

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- Xilinx PG150 (MIG 7 Series), PG065 (Clocking Wizard), PG058 (BMG), PG247 (SmartConnect)
- The KC705 board user guide, particularly the DDR3 and clock topology
- Deliverable: a 3-page decision memo recommending an IP and a configuration for each function, with the licensing and portability implications stated

### DELIVERABLES

- fpga/ip/ with a reproducible Tcl configuration script per IP
- The decision memo and a resource/timing table
- A README describing how to regenerate every IP from scratch

### DEFINITION OF DONE

- Each IP configuration builds from its Tcl script with no GUI steps
- Measured resource and f\_max numbers recorded for each
- The WP16 owner accepts the memo as the basis for board bring-up

### MILESTONES

- W1–2** Reading the product guides and the board user guide
- W3** MIG evaluation and test project
- W4** Clocking, BRAM and interconnect IP
- W5** Debug cores
- W6** Decision memo and Tcl scripting

**Start here:** Spec §29 (FPGA implementation), §30 (area/timing budgets), PROJECTS.md WP16

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Docs & Research

NFR-7 requires every module to carry a README stating its interface contract. Making that true — and building the gate that keeps it true — is how this project survives the cohort turnover that is certain in a student lab. You will read more of the codebase than almost anyone, which is the real reward here.

- A README template: purpose, ports, parameters, timing assumptions, reset behaviour, owner, backup
- A README for every module that lacks one, written by reading the RTL and interviewing the owner
- A handover-note section in each module README: current state, known issues, what to do next
- A docs index page linking every module README
- A list of the places where the RTL and the specification disagree, filed as issues

- The Definition of done in PROJECTS.md — what every project must leave behind
- Two well-documented open-source hardware repos, for what good looks like
- Deliverable: the README template itself, reviewed and agreed before the sweep starts

- The template, and a README in every module directory
- The docs index and the RTL-versus-spec discrepancy list

- The M-10 CI documentation gate passes across the whole tree
- Every README names an owner and a backup
- Every discrepancy found is filed as an issue, not silently fixed

|             |                                       |
|-------------|---------------------------------------|
| <b>W1</b>   | Template design and review            |
| <b>W2–3</b> | The sweep, module by module           |
| <b>W4</b>   | Index, discrepancy list, issue filing |

**Start here:** NFR-7, spec Appendix D

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Docs & Research

Read CVA6, Ibex and Rocket — their structure, not their code — and present what each chose and why. Phase 0 requires this before S1-Core RTL is written, because the cheapest way to avoid a design mistake is to find someone who already made it. Three contributors, one core each, 45 minutes of presentation each.

- One core per person: pipeline structure, hazard strategy, memory interface, privilege support
- A comparison table across all three: stages, issue width, branch prediction, cache structure, extension mechanism, verification approach
- For each core, the single design decision you think is most worth copying, and why
- For each, the decision you think is worth avoiding, and why
- A 45-minute presentation per core, delivered to the lab

- The CVA6, Ibex and Rocket Chip papers and their documentation
- MEDS-S1 SCOPE\_CONTRACT §3 — read it after your review and check whether our deferrals still look right to you
- Deliverable: a 6–8 page comparison report, plus three presentations

- The comparison report in docs/reviews/
- Three recorded or delivered presentations
- A short list of proposed changes to the MEDS-S1 design, if you find any

- All three presentations delivered to the lab
- The comparison table is complete with no unfilled cells
- At least one concrete, argued recommendation reaches the Phase-0 design review

|           |                                                      |
|-----------|------------------------------------------------------|
| <b>W1</b> | Assign cores; read the papers                        |
| <b>W2</b> | Read the documentation and structure; draft sections |
| <b>W3</b> | Comparison table and report                          |
| <b>W4</b> | Presentations                                        |

**Start here:** SCOPE CONTRACT §3, ADDENDUM.md

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Docs & Research

Survey how the literature attaches accelerators to processors, and — more usefully — how honestly it measures them. The recurring failure in accelerator papers is a block that was never attached to anything, reporting a kernel-only speedup. Your job is to characterise that failure precisely enough that MEDS-S1's methodology can avoid it.

- A survey of coupling mechanisms: custom instructions, coprocessor interfaces, memory-mapped accelerators, and where each is used in practice
- A read of the CV-X-IF specification against our MXIF profile in INTERFACES.md
- A review of 8–10 recent edge-AI accelerator papers, recording exactly what each measured
- A tally: how many report end-to-end results, how many report kernel-only, how many report energy as measured versus modelled
- A recommendation on what MEDS-S1's §34 reporting table should require

- OpenHW CV-X-IF specification; MEDS-S1 INTERFACES.md §1 and §4
- Spec §21 (choosing a coupling mechanism), §31 and §34
- Deliverable: a 6-page review with the measurement tally as its central table

- The review in docs/reviews/
- A one-page recommendation on the §34 reporting requirements
- A 30-minute presentation to the lab

- At least eight papers reviewed with the measurement tally completed for each
- The recommendation is specific enough to act on, not general advice
- Presented to the lab; the MEDS-V team has read it

|             |                                        |
|-------------|----------------------------------------|
| <b>W1</b>   | Coupling mechanisms and CV-X-IF        |
| <b>W2–3</b> | Paper survey and the measurement tally |
| <b>W4</b>   | Write-up                               |
| <b>W5</b>   | Recommendation and presentation        |

NOTES · QUESTIONS FOR YOUR MENTOR

Mentee Verification

|           |          |          |                                                              |
|-----------|----------|----------|--------------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                                |
| 2         | 6 weeks  | P1       | rv-workshop completed, with your own single-cycle RV32I core |

## WHAT YOU WILL BUILD

- ## LITERATURE REVIEW & THE NOTE IT PRODUCES

- ## DELIVERABLES

- ## DEFINITION OF DONE

- ## MILESTONES

- Start here:** ONBOARDING.md Week 1, NFR-9, INTERFACES.md

NOTES · QUESTIONS FOR YOUR MENTOR

Mentor Core

|           |          |          |                                            |
|-----------|----------|----------|--------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                              |
| 2         | 10 weeks | P0       | Solid SystemVerilog; pipeline fundamentals |

Own the front of the pipeline: program counter generation, static BTBN prediction, the `fetch_req/fetch_rsp` interface and compressed-instruction expansion. The interface is the point — it is specified so the predictor can be swapped later without touching the pipeline, which turns a v2 branch-predictor study into a clean, measurable project with a baseline.

## WHAT YOU WILL BUILD

- PC generation with redirect from branch resolution, traps and debug entry
- Static BTfN prediction: backward taken, forward not taken
- The fetch\_req/fetch\_rsp interface exactly as specified, with the predictor behind it
- C-extension expansion to 32-bit encodings before decode
- Misaligned fetch handling and the instruction-access-fault path
- A unit testbench covering redirect, expansion and every fetch fault case

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- Spec §6 and §7.1; INTERFACES.md for the fetch interface contract
- The frontend structure of Ibex and CVA6 — see M-14's review if it has landed
- Deliverable: a 2-page design note on the redirect path and its timing, presented at a design review before RTL is written

## DELIVERABLES

- rtl/core/frontend/ with the RTL and its unit testbench
- The design note and a module README stating the interface contract
- A measured baseline branch-prediction accuracy on the benchmark suite

## DEFINITION OF DONE

- Design review passed before RTL; testbench green in CI; lint clean
- Fetch interface matches INTERFACES.md with no local amendments
- Baseline prediction accuracy published, so a v2 predictor has something to beat

## MILESTONES

- |             |                                                  |
|-------------|--------------------------------------------------|
| <b>W1–2</b> | Reading, design note, design review              |
| <b>W3–5</b> | PC, redirect and fetch interface RTL             |
| <b>W6–7</b> | BTFN and C-expansion                             |
| <b>W8–9</b> | Unit testbench and fault cases                   |
| <b>W10</b>  | Integration, baseline measurement, documentation |

**Start here:** Spec §6, §7.1, INTERFACES.md, SCOPE CONTRACT §2.1, PROJECTS.md WP2

NOTES · QUESTIONS FOR YOUR MENTOR

## Core Backend — Decode, Register File, ALU and Hazard Control

Mentor Core

|           |          |          |                                                              |
|-----------|----------|----------|--------------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                                |
| 2–3       | 12 weeks | P0       | Strong SystemVerilog; comfortable with the ISA specification |

Own the middle of the pipeline — decode, the register file, the ALU, the full forwarding network and the hazard control that ties them together. This is the largest core package and everything downstream depends on it, including the completion buffer that sits on the project's critical path.

### WHAT YOU WILL BUILD

- Decoder for RV64IMAC\_Zicsr\_Zifencei, generated from a machine-readable instruction list
- Register file with the required read ports and write arbitration
- ALU, branch resolution and the address-generation unit
- Full EX->EX and MEM->EX forwarding per spec §8.1, and single-cycle load-use stall
- The hazard unit implementing every row of the §8.2 hazard table
- The multi-cycle unit dispatch port that MUL, DIV and MXIF all attach to

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- Spec §7 and §8 in full; the RISC-V unprivileged specification for the instruction set
- Spec §9 — read it early, because the completion buffer constrains your write-back path
- Deliverable: a 3-page microarchitecture note including the decode table format and the forwarding matrix, presented at a design review before RTL

### DELIVERABLES

- rtl/core/backend/ with RTL, the generated decoder and unit testbenches
- The microarchitecture note and module READMEs
- Coordination with M-06, who verifies your ALU and forwarding paths

### DEFINITION OF DONE

- Design review passed before RTL
- M-06's forwarding coverage reaches 100% against your implementation
- RV64I architectural tests pass against Sail once the frontend and CSR are present

### MILESTONES

- W1–2** Reading, microarchitecture note, design review
- W3–5** Decoder and register file
- W6–8** ALU, branch, AGU
- W9–10** Forwarding and hazard unit
- W11–12** Multi-cycle dispatch port, integration, documentation

**Start here:** Spec §7, §8, §9, PROJECTS.md WP3 (critical path)

NOTES · QUESTIONS FOR YOUR MENTOR

## CSR File, Trap Architecture and Performance Counters

Mentor Core

| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                                     |
|-----------|----------|----------|-------------------------------------------------------------------|
| 2         | 12 weeks | P0       | Careful reading of the privileged specification;<br>SystemVerilog |

Own privilege. The CSR file, the M-mode trap architecture, the privilege state machine and the performance counters. S-mode CSRs and delegation are architected now even though behaviour is stubbed until Phase 5 — doing that here costs weeks and saves rewriting the trap logic and access-control matrix during Linux bring-up.

### WHAT YOU WILL BUILD

- A generated CSR file — the access-control matrix comes from a table, never hand-written
- Full M-mode trap architecture: mstatus, mtvec, mepc, mcause, mtval, mie, mip
- S-mode CSRs, medeleg and mideleg present and architected; translation stubbed
- Privilege FSM including DebugMode, with dcsr, dpc and dscratch
- mcycle, minstret and mhpcounter3–15 with programmable event selectors and mcountinhibit
- A unit testbench covering every CSR access permission and every trap cause

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The RISC-V privileged specification, machine and supervisor levels, in full
- Spec §10, §12, §13 and Appendix C (the CSR list)
- Deliverable: the complete CSR access-control matrix as a reviewed table, before any RTL — this table is the design

### DELIVERABLES

- rtl/core/csr/ with the generator, the RTL and the unit testbench
- The access-control matrix and the counter event list
- Coordination with M-09, who consumes your counters from software

### DEFINITION OF DONE

- Every CSR in Appendix C is implemented or explicitly and correctly absent
- Illegal access to every CSR traps correctly, proven by testbench
- Architectural tests for Zicsr pass against Sail

### MILESTONES

- W1–2** Reading; the access-control matrix; design review
- W3–5** CSR file and generator
- W6–8** Trap architecture and privilege FSM
- W9–10** Performance counters and event selectors
- W11–12** Testbench, arch-test bring-up, documentation

**Start here:** Spec §10, §12, §13, Appendix C, SCOPE\_CONTRACT §4, PROJECTS.md WP4

NOTES · QUESTIONS FOR YOUR MENTOR

**Mentor** **Fabric & Periph**

Own the interconnect. A 256-bit backbone with 64-bit core ports and a 32-bit AXI4-Lite peripheral subtree, sized against a declared bandwidth budget rather than a guess. Everything in the SoC talks through what you build, and the width choice is what stops MEDS-V being throttled to scalar bandwidth.

- Crossbar configuration using pulp-platform/axi, with our master and slave topology
- Upsizers and downsizers between the 64-bit, 256-bit and 32-bit domains
- Address decode driven from soc.yaml, never hand-maintained
- The AXI4-Lite bridge to the peripheral subtree
- The uncached DRAM alias (spec §18.4) and its PMA handling
- Bandwidth validation against the §18.3 budget, measured in simulation

- The AMBA AXI4 specification — the ordering and outstanding-transaction rules especially
- Spec §18 in full, particularly §18.3 (the bandwidth budget) and §18.4
- Deliverable: a 2-page note validating or challenging the §18.3 bandwidth budget with your own arithmetic, presented before RTL

- rtl/fabric/ with the crossbar configuration, adapters and decode logic
- A fabric testbench with traffic generators at the declared bandwidths
- The bandwidth validation note and module READMEs

- All four named configurations elaborate in CI
- Measured fabric throughput meets the \$18.3 budget in simulation
- Address decode is generated; no hand-written duplicate of the memory map exists

|              |                                                         |
|--------------|---------------------------------------------------------|
| <b>W1–2</b>  | AXI reading; bandwidth validation note; design review   |
| <b>W3–5</b>  | Crossbar configuration and topology                     |
| <b>W6–7</b>  | Width adapters                                          |
| <b>W8</b>    | Address decode from soc.yaml                            |
| <b>W9–10</b> | Traffic testbench, bandwidth measurement, documentation |

NOTES · QUESTIONS FOR YOUR MENTOR

**Mentor** **Fabric & Periph**

Own how the machine is interrupted. CLINT provides the timer and software interrupts the privileged specification requires; PLIC multiplexes every peripheral and accelerator interrupt into the core. All sources are level-sensitive, which is a deliberate simplification that removes a whole class of lost-interrupt bug.

- CLINT: mtime, mtimecmp per hart, msip, with the correct clock source
- PLIC: priority, pending, enable and threshold registers, claim and complete flow
- Level-sensitive handling for all sources, including accelerator IRQ lines from the sockets
- Interrupt-source allocation driven from soc.yaml
- A testbench covering priority ordering, threshold masking, and the claim/complete race

- The RISC-V PLIC specification and the CLINT model in the privileged specification
- Spec §24 (peripherals and interrupts) and §20.2 (socket IRQ behaviour)
- Deliverable: a 1-page note on the claim/complete race and how your design avoids losing or double-servicing an interrupt

- rtl/peripherals/clint/ and rtl/peripherals/plic/ with unit testbenches
- soc.yaml interrupt allocation schema and generated headers
- The race-condition note and module READMEs

- Priority, threshold and claim/complete all proven by directed testbench
- A timer interrupt reaches a C handler on the Verilator board
- Accelerator socket IRQ lines route correctly through to the core

|             |                                     |
|-------------|-------------------------------------|
| <b>W1</b>   | Reading and the race note           |
| <b>W2-3</b> | CLINT                               |
| <b>W4-6</b> | PLIC                                |
| <b>W7</b>   | soc.yaml integration                |
| <b>W8</b>   | Testbench closure and documentation |

NOTES · QUESTIONS FOR YOUR MENTOR

**Mentor** **Software & BSP**

|           |          |          |                                            |
|-----------|----------|----------|--------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                              |
| 2         | 10 weeks | P0       | C, assembly, linker scripts, build systems |

## WHAT YOU WILL BUILD

- ## LITERATURE REVIEW & THE NOTE IT PRODUCES

- ## DELIVERABLES

- ## DEFINITION OF DONE

- ## MILESTONES

|             |                                 |
|-------------|---------------------------------|
| <b>W1-2</b> | Reading and the HAL API note    |
| <b>W3-4</b> | crt0 and the linker integration |
| <b>W5-6</b> | newlib, syscalls, malloc        |
| <b>W7-8</b> | HAL and printf paths            |
| <b>W9</b>   | make run across boards          |
| <b>W10</b>  | Onboarding test and revision    |

NOTES · QUESTIONS FOR YOUR MENTOR

Mentor Verification

|           |          |          |                                              |
|-----------|----------|----------|----------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                |
| 2         | 8 weeks  | P0       | Python, CI systems, patience with toolchains |

Own the gate. RISCOF against Sail on every merge, wired into CI so that a second-year student learns within twenty minutes whether their change broke the ISA. This has to be green before the RTL exists — a verification harness built after the core is a harness that never gets built.

- RISCOF with Sail as the golden model, running green on an empty core in Phase 0
- The full arch-test suite wired into per-PR CI within the 20-minute budget (NFR-3)
- Test selection per configuration, since the four configs implement different subsets
- Result reporting into the evidence-bundle format
- The nightly full-suite job, separate from the fast per-PR job

- RISCOF documentation; the Sail RISC-V model and how to build it
- Spec §28 (verification architecture), particularly layers 2 and 3, and §28.5 (CI policy)
- Deliverable: a 1-page plan for keeping the per-PR job under 20 minutes as the suite grows, agreed before you build

- verif/riscov/ with plugins, configuration and the CI job definitions
- The evidence-bundle report generator
- Coordination with M-11, who triages the failures your harness finds

- CI green on an empty core before Phase 1 starts — the Phase-0 exit criterion
- Per-PR job completes within 20 minutes; nightly job runs the full suite
- Nothing can be merged with the gate red

|             |                                       |
|-------------|---------------------------------------|
| <b>W1</b>   | Reading and the CI budget plan        |
| <b>W2–3</b> | RISCOF and Sail running locally       |
| <b>W4–5</b> | CI integration and the runner         |
| <b>W6</b>   | Per-config test selection             |
| <b>W7–8</b> | Reporting, nightly job, documentation |

**Start here:** Spec §28, §28.5, §28.6, NFR-3, PROJECTS.md WP12

NOTES · QUESTIONS FOR YOUR MENTOR

Mentor

|           |          |          |                                                            |
|-----------|----------|----------|------------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                              |
| 2-3       | 12 weeks | P1       | Vivado, FPGA flow, timing constraints; M-12's memo in hand |

## WHAT YOU WILL BUILD

- The four-file board port: board.yaml, board.xdc, board\_top.sv, openocd.cfg
- Clock and reset topology, PLL configuration, and the CDC into the peripheral domain
- DDR3 via MIG, using M-12's evaluated configuration, through to calibration success
- The bring-up sequence in spec §29.2, executed in order, with each step recorded
- Timing closure at 50 MHz minimum (NFR-1), with the report published
- An out-of-context synthesis flow so accelerator work does not rebuild the whole SoC

- Spec §29 (FPGA implementation) and §30 (area, timing and power budgets)
- M-12's Xilinx IP decision memo; the KC705 board user guide
- Deliverable: a bring-up log kept from day one, recording every step of §29.2 and every failure — this becomes the guide for the next board

- `fpga/boards/kc705/` with the four files and the build scripts
- The bring-up log and the published timing and utilisation reports
- The out-of-context synthesis flow and its documentation

- Every step of the \$29.2 bring-up order passed, in order, and recorded
- Timing met at the declared frequency; utilisation report published
- A C program prints over UART from the FPGA, not from simulation

|               |                                              |
|---------------|----------------------------------------------|
| <b>W1-2</b>   | Reading, M-12 handover, constraint authoring |
| <b>W3-4</b>   | Bitstream, clocks, LED, UART bring-up        |
| <b>W5-7</b>   | DDR3 and MIG calibration                     |
| <b>W8-10</b>  | Full SoC integration and timing closure      |
| <b>W11-12</b> | Out-of-context flow, reports, documentation  |

NOTES · QUESTIONS FOR YOUR MENTOR

Graduate RA Core

|           |          |                    |                                                                 |
|-----------|----------|--------------------|-----------------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY           | PREREQUISITES                                                   |
| 1–2       | 12 weeks | P0 — critical path | Strong microarchitecture; formal or assertion experience valued |

## WHAT YOU WILL BUILD

- An 8-entry completion buffer with the entry format of spec §9.1
- Retire rules per §9.2, preserving precise exceptions including for offloaded instructions
- The MXIF port implementing MXIF-1.0: non-speculative issue, two-phase completion
- Offload at the retire pointer, so no coprocessor side effect is ever speculative
- SVA liveness properties per §9.4, running from day one rather than added at the end
- A conformance testbench driving every MXIF handshake case, including rejection and faults

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- Spec §9 in full, especially §9.3 (why eight entries) and §9.4 (the deadlock argument)
- INTERFACES.md §1 in full — MXIF-1.0 is normative and you may not amend it locally
- The OpenHW CV-X-IF specification, and our profile's deviations from it
- Deliverable: a written deadlock argument for your implementation, independent of the spec's, presented at a design review before RTL

## DELIVERABLES

- `rtl/core/completion/` with the buffer, retire logic and the MXIF port
- `verif/conformance/tb_mxif_conformance.sv` — a reusable asset that outlives every coprocessor
- The SVA property set and the deadlock argument

## DEFINITION OF DONE

- Liveness properties pass under bounded model checking
- Conformance testbench covers every handshake case in INTERFACES.md §1
- MEDS-V attaches over this port with zero changes to core RTL — the real test

## MILESTONES

- |               |                                              |
|---------------|----------------------------------------------|
| <b>W1-3</b>   | Reading, deadlock argument, design review    |
| <b>W4-6</b>   | Completion buffer and retire                 |
| <b>W7-9</b>   | MXIF port and two-phase completion           |
| <b>W10-11</b> | Conformance testbench and SVA                |
| <b>W12</b>    | MEDS-V attachment trial with the MEDS-V team |

**Start here:** Spec: §9, §19: INTERFACES.md §1: PROJECTS.md WP5

NOTES · QUESTIONS FOR YOUR MENTOR

Graduate RA      Memory

| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                   |
|-----------|----------|----------|-------------------------------------------------|
| 1–2       | 12 weeks | P0       | Strong microarchitecture; memory-model literacy |

## WHAT YOU WILL BUILD

- LSU with the full set of load and store widths, sign extension and alignment handling
- A store buffer with correct load-forwarding from pending stores
- 16 PMP regions with standard granularity and locking
- The PMA check unit — coordinate with M-05, who writes its testbench
- AMO and LR/SC per the A extension, including the reservation set rules
- The scalar/coprocessor memory interlock of INTERFACES.md §1.5, and its measured cost

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- Spec §14 (load/store unit) and §11 (PMP and PMA)
- The RISC-V memory model chapter, and the A-extension specification
- Deliverable: a 2-page note on the interlock — what it costs, how you will measure it, and what a v2 disambiguation scheme would have to beat

## DELIVERABLES

- rtl/core/lsu/ with the LSU, store buffer, PMP and PMA units
- Unit testbenches, plus support for M-05's PMA testbench
- The interlock cost measurement, published

## DEFINITION OF DONE

- A-extension architectural tests pass against Sail
- PMP tests pass; M-05's PMA coverage closes against your implementation
- The interlock cost is measured on the benchmark suite and reported honestly

## MILESTONES

- |               |                                                  |
|---------------|--------------------------------------------------|
| <b>W1–2</b>   | Reading, interlock note, design review           |
| <b>W3–5</b>   | LSU and store buffer                             |
| <b>W6–7</b>   | PMP and PMA                                      |
| <b>W8–10</b>  | Atomics and reservations                         |
| <b>W11–12</b> | Interlock measurement, arch-tests, documentation |

**Start here:** Spec §11. §14: INTERFACES.md §1.5: SCOPE CONTRACT §3: PROJECTS.md WP6

NOTES · QUESTIONS FOR YOUR MENTOR

Graduate RA Memory

Build the caches and — just as importantly — the SRAM wrapper that keeps this design ASIC-clean. Blocking, 2-way, write-back with a write buffer: correct and adequate rather than clever, because non-blocking is a measurable v2 optimisation and a better project once there is a baseline to beat.

- Parameterised I\$ and D\$: 2-way, 64-byte lines, configurable size
- Write-back D\$ with a write buffer, and the coherence-free software model via Zicbom
- `cbo.clean`, `cbo.flush` and `cbo.inval`, with correct interaction with the store buffer
- `meds_sram_wrapper` — every memory in the design goes through it, no exceptions (NFR-5)
- Cache testbenches: hit, miss, eviction, write-back ordering, `cbo` semantics
- Measured hit rates on the benchmark suite, published as the v1 baseline

- Spec §15 (caches) and §17 (the SRAM wrapper)
- The Zicbom specification; SCOPE\_CONTRACT §3 on why coherence is deferred
- Deliverable: a 2-page note on the Zicbom software contract — what software must do around a DMA buffer — which becomes the driver template's documentation

- rtl/core/cache/ and rtl/common/meds\_sram\_wrapper.sv with unit testbenches
- The Zicbom software contract note, handed to the BSP team
- Baseline hit-rate measurements across the workload suite

- No memory anywhere in the design is instantiated outside `meds_sram_wrapper`
- cbo semantics proven correct against a DMA-style testbench
- Hit rates published, giving a v2 cache project a baseline

|               |                                              |
|---------------|----------------------------------------------|
| <b>W1–2</b>   | Reading, Zicbom contract note, design review |
| <b>W3–4</b>   | SRAM wrapper and the memory abstraction      |
| <b>W5–7</b>   | IS                                           |
| <b>W8–10</b>  | D\$ and write buffer                         |
| <b>W11–12</b> | Zicbom, testbenches, baseline measurement    |

NOTES · QUESTIONS FOR YOUR MENTOR

# The SoC Generator — soc.yaml to RTL, Linker, Headers and DTS

Graduate RA    FPGA & Tools

|           |          |          |                                                   |
|-----------|----------|----------|---------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                     |
| 1-2       | 12 weeks | P0       | Strong Python; templating; SystemVerilog literacy |

Build the single source of truth. One soc.yaml generates the SoC top level, the linker script, the C headers, the device tree, the memory-map documentation, the Verilator top, the OpenOCD config and the PMA decode logic. The failure this prevents — a memory map correct in the RTL and wrong in the device tree — is otherwise inevitable and brutal to debug.

## WHAT YOU WILL BUILD

- The soc.yaml schema, versioned, with validation and clear error messages
- Generators for: SoC top-level RTL, link.ld, C headers, .dtsi, memory\_map.md
- Generators for the Verilator top, the OpenOCD configuration and the PMA decode logic
- Golden tests: a known soc.yaml produces byte-identical outputs, checked in CI
- The peripheral-addition path documented well enough that a mentee can use it unaided

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- Spec §26 (the SoC generator) and Appendix B (default memory map)
- One existing SoC generator — Chipyard, LiteX or similar — and what it gets right and wrong
- Deliverable: a 2-page schema design note, reviewed before implementation, because the schema is an interface and changing it later is expensive

## DELIVERABLES

- generator/ with the schema, generators, templates and golden tests
- The schema note and a contributor guide for adding a peripheral
- Generated outputs wired into the build so no hand-maintained duplicates remain

## DEFINITION OF DONE

- Every artefact in spec §26 is generated; no hand-maintained duplicate exists anywhere
- Golden tests pass in CI
- A contributor who has only done rv-workshop adds a peripheral and sees it from C, in under a day, following the documentation alone (NFR-9)

## MILESTONES

- W1-2    Reading, schema design note, review
- W3-5    RTL and linker-script generation
- W6-7    Headers, DTS, documentation generation
- W8-9    Verilator top, OpenOCD, PMA decode
- W10-11    Golden tests and CI
- W12    Contributor guide and the NFR-9 test

Start here: Spec §26, Appendix B; NFR-9; PROJECTS.md WP10

NOTES · QUESTIONS FOR YOUR MENTOR

# RVFI Trace Port and the Spike Co-simulation Harness

Graduate RA Verification

|           |          |          |                                               |
|-----------|----------|----------|-----------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                 |
| 1-2       | 10 weeks | P0       | SystemVerilog and C++; DPI; debugging stamina |

The highest-value verification investment in the project. A directed test costs twenty minutes and covers one case; this harness costs three days and covers every case any program ever exercises, forever. Without it the only signal is "the answer is wrong"; with it the signal names the instruction, the PC, the register and the expected value.

## WHAT YOU WILL BUILD

- An RVFI port driven from the first pipeline commit, per INTERFACES.md §5
- The MEDS rvfi\_v\_\* extension group for vector state, including byte-granular write masks
- A Spike co-simulation harness comparing every retired instruction over DPI
- Divergence reporting that names instruction number, PC, mnemonic, and expected versus actual
- CI integration on every PR, within the NFR-3 budget
- Trace capture on failure, so a divergence can be reproduced offline

## LITERATURE REVIEW & THE NOTE IT PRODUCES

- The RVFI specification from riscv-formal; INTERFACES.md §5
- Spec §28.1 and §28.2 — why co-simulation dominates, and the RVFI contract
- The MEDS-V book's co-simulation chapter, which solved this problem once already
- Deliverable: a 2-page note on how vector state is checked, since comparing final register contents is exactly what misses undisturbed-tail bugs

## DELIVERABLES

- rtl/core/rvfi/ and verif/cosim/ with the harness and the DPI layer
- CI job running co-simulation on every PR
- The vector-checking note, and documentation on reading a divergence report

## DEFINITION OF DONE

- Every retired instruction is compared against Spike across the whole benchmark suite
- A deliberately injected bug is caught and correctly localised by the report
- Runs on every PR within the CI budget

## MILESTONES

- W1-2**
- Reading, vector-checking note, design review
- W3-4**
- RVFI port
- W5-7**
- Spike harness and the DPI layer
- W8**
- Divergence reporting
- W9-10**
- CI integration, fault injection testing, documentation

Start here: Spec §28.1, §28.2; INTERFACES.md §5; PROJECTS.md WP11

NOTES · QUESTIONS FOR YOUR MENTOR

## Debug Module, JTAG DTM, OpenOCD and Semihosting

Graduate RA

FPGA &amp; Tools

|           |          |          |                                                       |
|-----------|----------|----------|-------------------------------------------------------|
| TEAM SIZE | DURATION | PRIORITY | PREREQUISITES                                         |
| 1–2       | 10 weeks | P0       | JTAG, GDB internals, C; T-03's privilege FSM in place |

Deliver the capability that unlocks the whole lab. Until an arbitrary ELF can be loaded and debugged over JTAG with no resynthesis, every software change costs a bitstream build and the platform is useful only to the people building it. This is the Phase-3 exit criterion, and everything else in Phase 3 is downstream of it.

### WHAT YOU WILL BUILD

- RISC-V Debug Module integration using `pulp-platform/riscv-dbg`, plus the JTAG DTM
- Halt, resume, single step, and two instruction-address triggers
- Abstract commands for register access, and system-bus access for memory
- OpenOCD configuration generated from `soc.yaml`, and a working GDB connection
- Semihosting: open, read, write, close and lseek proxied to the host filesystem
- Debug entry and exit correctness against the privilege FSM, proven by testbench

### LITERATURE REVIEW & THE NOTE IT PRODUCES

- The RISC-V External Debug Support specification
- Spec §13 (debug) and §27.2 (semihosting)
- Deliverable: a 2-page note on debug-mode entry and its interaction with traps and the completion buffer, agreed with the R-01 and T-03 owners before RTL

### DELIVERABLES

- `rtl/debug/` with the DM and DTM integration and its testbench
- Generated OpenOCD configuration and the GDB workflow documentation
- Semihosting implementation in the BSP, with the host-side proxy

### DEFINITION OF DONE

- `openocd` and `gdb` load and debug an arbitrary ELF over JTAG with no resynthesis — the Phase-3 exit criterion
- Semihosting moves a multi-megabyte file to the target without an SD card
- Halt, step, breakpoint and register access all proven on hardware

### MILESTONES

- W1–2** Reading, debug-entry note, design review
- W3–5** DM and DTM integration
- W6–7** OpenOCD and GDB bring-up
- W8–9** Semihosting
- W10** Hardware validation and documentation

**Start here:** Spec §13, §27.2; RISC-V Debug Spec; PROJECTS.md WP15

NOTES · QUESTIONS FOR YOUR MENTOR

Graduate RA Memory

Build the loosely-coupled attachment point most ML thesis projects will use, and the conformance testbench that keeps it honest. Clock-domain crossing lives in the socket so accelerator authors never write synchronisers. This is the module that turns MEDS-S1 from a processor into a platform.

- meds\_s1\_accel\_socket with AXI4-Lite config window, AXI4 DMA master and CDC, per spec §20.1
- The mandatory register map of §20.2 — including PERF\_CYCLES and PERF\_STALLS at fixed offsets
- Level-sensitive IRQ synchronisation into the PLIC
- ASYNC=0/1 support so an accelerator can run in its own clock domain safely
- `verif/conformance/tb_socket_conformance.sv` covering the register map, IRQ assert and clear, abort, error reporting, and DMA to both legal and illegal regions
- A driver template with the cache maintenance already correct, since that is what students miss

- Spec §20 in full, §21 (choosing a coupling mechanism), §22 (data movement patterns)
- INTERFACES.md §4 — the socket interface is normative
- Deliverable: a 2-page note on the CDC strategy and why accelerator authors are forbidden from writing their own synchronisers (NFR-6)

- rtl/socket/ with the socket RTL and its parameterisation
- The socket conformance testbench — a reusable asset that outlives every accelerator
- The driver template and the accelerator author's guide

- The conformance testbench covers every case in INTERFACES.md §4
- Two trivial example accelerators attach and pass conformance
- An accelerator author following the guide alone attaches a stub in under a day

|             |                                        |
|-------------|----------------------------------------|
| <b>W1–2</b> | Reading, CDC note, design review       |
| <b>W3–5</b> | Socket RTL, register window, DMA path  |
| <b>W6–7</b> | CDC and IRQ synchronisation            |
| <b>W8–9</b> | Conformance testbench                  |
| <b>W10</b>  | Driver template and the author's guide |

NOTES · QUESTIONS FOR YOUR MENTOR